LECTURE-4

(Be careful about the typos)

Escape Velocity: $m\dfrac{dv}{dt} = -\dfrac{GMm}{x^2}$, where G is the universal gravitational constant $G = 6.67 \times 10^{11}\ m^3\ Kg^{-1}s^{-2}$, r is the distance from the center of gravity. M and m are mass of the earth and particle respectively.

Solution:

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Solution:

$$\begin{aligned}
\frac{dv}{dt} &= -\frac{GM}{x^2} \quad \Rightarrow \quad \frac{dv}{dx}\frac{dx}{dt} = -\frac{GM}{x^2} \\
\Rightarrow \quad v\frac{dv}{dx} &= -\frac{GM}{x^2} \\
\Rightarrow \quad \int_{v_0}^{v(t)} vdv &= -\int_{x_0}^{x(t)} \frac{GM}{x^2} \\
\Rightarrow \quad \frac{v^2(t)}{2} - \frac{v_0^2}{2} &= \frac{GM}{x(t)} - \frac{GM}{x_0}
\end{aligned}$$

Since we are interested in escape velocity so as $t \to \infty$, $x(t) \to \infty$.

$$
\begin{aligned}
\frac{v_0^2}{2} &= \frac{v^2(t)}{2} + \frac{GM}{x_0} \\
v_{es}^2 &= \frac{2GM}{x_0} \quad [\text{ we can take } v = 0 \text{ for escape velocity}] \\
v_{es}^2 &= \frac{2R^2 g}{R} \quad [\text{ On earth surface } x_0 = R \text{ and } GM = R^2 g] \\
v_{es} &= \sqrt{2gR}
\end{aligned}
$$

For earth $R = 6400$ kilometer $= 6.4 \times 10^6$ meter.
$M = 5.97219 \times 10^{24} kg$, $g = 9.8\ m/s^2$ and
$G = 6.67 \times 10^{-11}\ m^3 kg^{-1} s^{-2}$.

Therefore $v_{es} = 11200\ m/s$.

Example: Find the minimum velocity of escape if the body is carried by a rocket and is projected from the rocket at a hight of 500 miles from the earth's surface.

Solution: The equation of motion of a body of mass m under the earth gravitational filed is $m\dfrac{d^2x}{dt^2} = -\dfrac{mMG}{x^2} = -\dfrac{mgR^2}{x^2}$.

Here G, M, g and R are the Gravitational constant, mass, gravitational force (gravity) and radius of the earth respectively.

$$\begin{aligned}
\therefore \quad \frac{d^2x}{dt^2} &= -\frac{gR^2}{x^2} \Rightarrow v\frac{dv}{dx} = -\frac{gR^2}{x^2} \\
\Rightarrow vdv &= -\frac{gR^2}{x^2}dx \Rightarrow \int vdv = -gR^2 \int \frac{1}{x^2}dx + C/2 \\
\Rightarrow v^2 &= \frac{2gR^2}{x} + C.
\end{aligned}$$

Let v_{500} be the velocity of projection of the body at a hight 500 miles from the earth surface i.e. when $x = R + 500$, $v = v_{500}$.

$$\begin{aligned} v^2 &= \frac{2gR^2}{x} + C \\ \therefore \quad v_{500}^2 &= \frac{2gR^2}{R+500} + C \Rightarrow C = v_{500}^2 - \frac{2gR^2}{R+500} \\ \Rightarrow \quad v^2 &= \frac{2gR^2}{x} + v_{500}^2 - \frac{2gR^2}{R+500} \end{aligned}$$

If $v_{500}^2 \geq \frac{2gR^2}{R+500}$, then $v^2 \geq \frac{2gR^2}{x} \geq 0$ for all values of x and the body escape from the earth's gravitational pull.

But if $v_{500}^2 < \frac{2gR^2}{R+500}$, say $v_{500}^2 = \frac{2gR^2}{R+500} - u$, where $u > 0$. then $v^2 = \frac{2gR^2}{x} - u$, which becomes zero for $x = \frac{2gr^2}{u}$ and the body would start falling down.

Thus the minimum velocity of escape at hight 500 miles from earth surface is $v_e = v_{500} = \sqrt{\frac{2gR^2}{R+500}} \simeq 6.55$ miles/second (Or, 10.5 km/s).

Note: $R = 6378.1km = 3960miles$;
$g = 9.8331m/s^2 = 32ft/s^2 = 0.0061miles/s^2$; $M = 5.97219^{24}kg$;
$G = 6.67428 \times 10^{-11}m^3/(kgs^2)$. 1 miles= 1.6093 km.

Radioactive decay

$$\frac{dm}{dt} = -km$$

Example: A body of mass m is dropped from rest from height h_0 in the earth's atmosphere. Assuming that the body falls vertically under the action of earth's gravity and neglecting all other effects except air resistance, which is proportional to its velocity, find its height h at any subsequent time.

Solution:

$$m\frac{dv}{dt} = mg - kv \quad \Rightarrow \quad \frac{dv}{dt} + \frac{k}{m}v = g$$

Therefore I.F. is $e^{\frac{k}{m}\int tdt} = e^{\frac{kt}{m}}$.
Hence, the solution of is

$$\begin{aligned} v &= e^{-\frac{kt}{m}}\left[\int ge^{\frac{kt}{m}}dx + c_1\right] \\ &= e^{-\frac{kt}{m}}\left[\frac{mg}{k}e^{\frac{kt}{m}} + c_1\right] \end{aligned}$$

Since, at $t = 0$, $v = 0$ so we have $c_1 = -\frac{mg}{k}$.

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Hence,

$$v = e^{-\frac{kt}{m}}\left[\frac{mg}{k}e^{\frac{kt}{m}} - \frac{mg}{k}\right] \Rightarrow \frac{dh}{dt} = \frac{mg}{k} - \frac{mg}{k}e^{-\frac{kt}{m}}$$

$$\Rightarrow \quad h = \frac{mgt}{k} + \frac{m^2 g}{k^2}e^{-\frac{kt}{m}} + c_2$$

Now, at $t = 0$, $h = h_0$. Hence $c_2 = h_0 - \frac{m^2 g}{k^2}$.

Therefore the solution becomes $h = h_0 + \frac{mg}{k}t - \frac{m^2 g}{k^2}\left(1 - e^{-\frac{kt}{m}}\right)$.